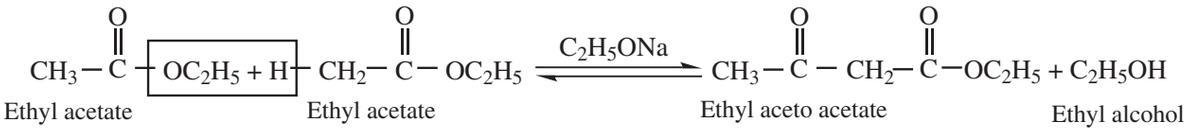
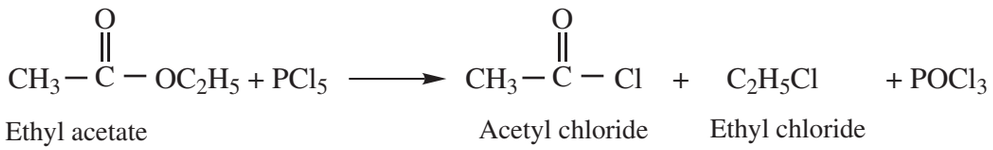
4. Claisen Condensation

Esters containing at least one ∝- hydrogen atom undergo self condensation in the presence of a strong base such as sodium ethoxide to form **β- keto ester.**



5. Reaction with PCl_5

Esters react with PCl_5 to give a mixture of acyl and alkyl chloride

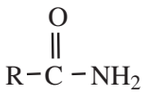


Evaluate yourself

Why is acid anhydride preferred to acyl chloride for carrying out acylation reactions ?

12.14.5 Acid Amides

Acid amides are derivatives of carboxylic acid in which the – OH part of carboxyl group has been replaced by – NH_2 group. The general formula of amides are given as follows.

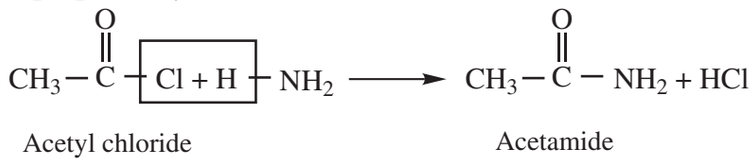


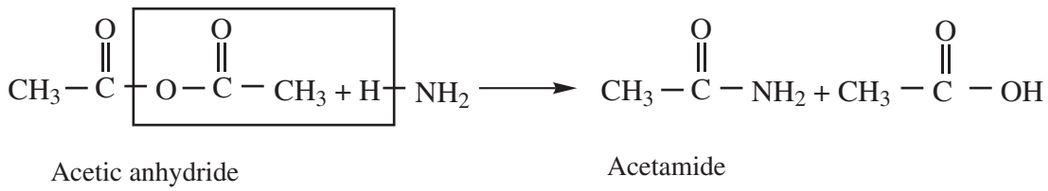
Now, we shall focus our attention mainly on the study of chemistry of acetamide.

Methods of Preparation

1. Ammonolysis of acid derivatives

Acid amides are prepared by the action of ammonia with acid chlorides or acid anhydrides.





2) Heating ammonium carboxylates

Ammonium salts of carboxylic acids (ammonium carboxylates) on heating, lose a molecule of water to form amides.



XII U12-Carbonyl Compounds.indd 186 11-01-2021 16:10:48